

Quantitative Comparisons



Numeric comparisons

- 1 Column A: 7×8 . Column B: $6 \times 9 + 2$. Compare the two columns: which is greater, are they equal, or is it impossible to determine?
 - 2 Column A: $123 + 456$. Column B: $700 - 120$. Compare the two columns.
 - 3 Column A: 9^2 . Column B: $8^2 + 15$. Compare the two columns.
 - 4 Column A: $(-3)^2$. Column B: -3^2 . Compare the two columns.
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Fraction and decimal comparisons

- 5 Column A: $\frac{3}{7}$. Column B: 0.43. Compare the two columns.
 - 6 Column A: $\frac{7}{12}$. Column B: 0.58. Compare the two columns.
 - 7 Column A: $2\frac{3}{8}$. Column B: $\frac{19}{8}$. Compare the two columns.
 - 8 Column A: 0.6. Column B: $\frac{5}{9}$. Compare the two columns.
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Percent comparisons

- 9 Column A: 25% of 160. Column B: 20% of 200. Compare the two columns.
 - 10 Column A: 30% of 90. Column B: 40% of 60. Compare the two columns.
 - 11 Column A: the result of increasing 100 by 10% twice in a row. Column B: the result of increasing 100 by 20% once. Compare the two columns.
 - 12 Column A: 15% of 80. Column B: 12% of 100. Compare the two columns.
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Exponent and root comparisons

- 13 Column A: $\sqrt{50}$. Column B: 7.1. Compare the two columns.
- 14 Column A: 2^{10} . Column B: 10^3 . Compare the two columns.

15 Column A: $\sqrt{0.09}$. Column B: 0.3. Compare the two columns.

16 Column A: 3^4 . Column B: 4^3 . Compare the two columns.

Algebraic comparisons at a fixed value

17 Given $x = 5$. Column A: $2x + 3$. Column B: $x + 8$. Compare the two columns.

18 Given $x = 4$. Column A: $(x + 2)^2$. Column B: $x^2 + 4x + 4$. Compare the two columns.

19 Given $n = -2$. Column A: n^2 . Column B: n^3 . Compare the two columns.

20 Given $x = 3$. Column A: $5x - 2$. Column B: $2(x + 5)$. Compare the two columns.

Comparisons requiring testing values

21 For any real number x : Column A: x . Column B: x^2 . Compare the two columns.

22 For any real number x : Column A: $x + 5$. Column B: $2x$. Compare the two columns.

23 For any real number x : Column A: $|x|$. Column B: x . Compare the two columns.

24 Given only that n is an integer: Column A: n^2 . Column B: 4. Compare the two columns.

25 For any real number x : Column A: $(x - 3)^2$. Column B: $x^2 - 6x + 9$. Compare the two columns.

Geometry comparisons

26 Column A: the perimeter of a square with side 6 cm. Column B: the perimeter of a rectangle with length 8 cm and width 3 cm. Compare the two columns.

27 Column A: the area of a circle with radius 4 cm (use $\pi \approx 3.14$). Column B: the area of a square with side 7 cm. Compare the two columns.

28 Column A: the sum of the interior angles of a pentagon. Column B: the sum of the interior angles of two triangles. Compare the two columns.

29 Column A: the hypotenuse of a right triangle with legs 5 and 12. Column B: 14. Compare the two columns.

Ratio and proportion comparisons

- 30** Column A: the value of x , if $x : 12 = 3 : 4$. Column B: 10. Compare the two columns.
- 31** A recipe uses flour and sugar in the ratio 3 : 1. Column A: the sugar needed for 240 g of flour. Column B: 90 g. Compare the two columns.
- 32** Column A: the sum of the parts in the simplified form of the ratio 18 : 24. Column B: 8. Compare the two columns.
- 33** Two numbers are in the ratio 5 : 3, and their sum is 96. Column A: the larger number. Column B: 61. Compare the two columns.

Data and probability comparisons

- 34** A data set is 4, 7, 7, 9, 13. Column A: the mean. Column B: the median. Compare the two columns.
- 35** Column A: the range of the data set 3, 8, 15, 22. Column B: 20. Compare the two columns.
- 36** A bag contains 3 red balls and 5 blue balls. Column A: the probability of drawing a red ball. Column B: the probability of drawing a blue ball. Compare the two columns.

Word-problem comparisons

- 37** Column A: the total cost of 4 items at 15 riyals each, after a 10% discount on the total. Column B: 55 riyals. Compare the two columns.
- 38** Ali's age is 3 years more than twice Sara's age. Sara is 8. Column A: Ali's age. Column B: 20. Compare the two columns.
- 39** Column A: the simple interest earned on 2,000 riyals at 5% per year for 3 years. Column B: 350 riyals. Compare the two columns.
- 40** A car travels 240 km using 20 liters of fuel. Column A: the distance travelled per liter. Column B: 12 km per liter. Compare the two columns.

Comparisons with inequalities and ranges

- 41 Given that $5 < x < 9$. Column A: x . Column B: 7. Compare the two columns.
- 42 Given that $x > 10$. Column A: x . Column B: 10. Compare the two columns.
- 43 Given that $0 < x < 1$. Column A: x . Column B: x^2 . Compare the two columns.
- 44 Given that $-5 \leq x \leq -1$. Column A: x^2 . Column B: 10. Compare the two columns.
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More geometry comparisons

- 45 Column A: the volume of a cube with edge 4 cm. Column B: the volume of a rectangular box 3 cm $\times$ 4 cm $\times$ 6 cm. Compare the two columns.
- 46 Column A: the circumference of a circle with radius 7 cm (use $\pi = \frac{22}{7}$). Column B: the perimeter of a square with side 11 cm. Compare the two columns.
- 47 Column A: the area of a triangle with base 10 and height 8. Column B: the area of a rectangle with length 9 and width 5. Compare the two columns.
- 48 Column A: the sum of the interior angles of a quadrilateral. Column B: three times the sum of the interior angles of a triangle. Compare the two columns.